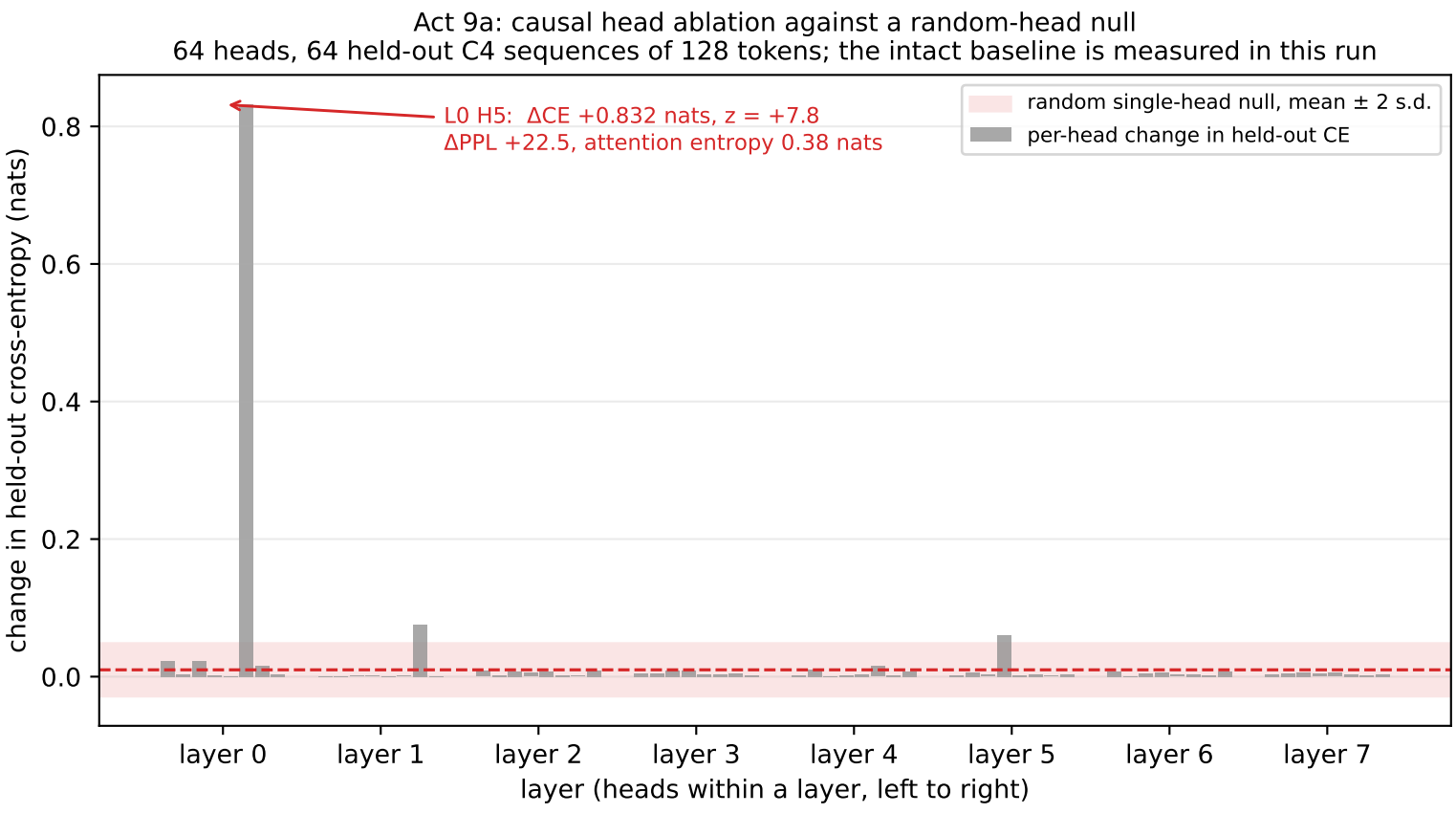
Act 9a: causal head ablation against a random-head null
64 heads, 64 held-out C4 sequences of 128 tokens; the intact baseline is measured in this run
L0 H5: ΔCE +0.832 nats, z = +7.8
ΔPPL +22.5, attention entropy 0.38 nats
random single-head null, mean ± 2 s.d.
per-head change in held-out CE
change in held-out cross-entropy (nats)
0.0
0.2
0.4
0.6
0.8
layer 0
layer 1
layer 2
layer 3
layer 4
layer 5
layer 6
layer 7
layer (heads within a layer, left to right)